The Orbital Response Network:

A Surgical Decomposition of Transformer Attention
into Shared Coupling and Per-Layer Response

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https://github.com/Percolation-Labs/orn

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Abstract

Prior work on transformer weight sharing has largely taken a blunt approach: tie whole blocks across layers (ALBERT, Universal Transformer), or tie everything and patch the gaps with LoRA (RRT). These trade quality for compression because they share what should not be shared. We argue for a different move. Pretrained transformer coupling matrices $W_Q^{(l)\top} W_K^{(l)}$ are spectrally correlated across layers even though their raw orientations are not, which isolates the coupling *rule* as the piece that is genuinely shared by training; layer-specific variation lives in the response, not in the coupling. The **Orbital Response Network** (ORN) acts on this measurement surgically: replace $2L$ per-layer W_Q, W_K with one shared $\mathbf{M} = AB^\top$, keep W_V, W_O, and the FFN per-layer, and let the per-layer rotation that used to sit inside W_Q, W_K be absorbed into the response heads via a standard orthogonal-gauge argument. The narrow claim of this paper is that the surgical substitution does not hurt: at 108M parameters on 8B tokens of FineWeb-Edu an ORN tracks a standard transformer at equal parameters across 1M–108M scales, a frozen $\mathbf{M}$ transferred across datasets recovers 98.6% of from-scratch quality, and the trained $\mathbf{M}$ crystallises to a 5–8 dimensional manifold consistent with what we measured on the pretrained transformers that motivated the architecture. A larger 605M variant combines shared $\mathbf{M}$ with a single wide shared FFN [20] and reaches val loss 2.82 on 3B tokens on a single A100. Published checkpoints, reproducible code, and a laptop-friendly diagnostics suite are at the repository.[1]

1 Introduction

A transformer attention head at layer l computes:

$$\mathrm{Attn}_l(X) = \mathrm{softmax}\left(\frac{X W_Q^{(l)} W_K^{(l)\top} X^\top}{\sqrt{d_h}}\right) X W_V^{(l)} \quad (1)$$

The effective coupling matrix $\mathbf{M}_l = W_Q^{(l)\top} W_K^{(l)} \in \mathbb{R}^{d\times d}$ determines which tokens attend to which. This matrix is learned independently at every layer, costing $\mathcal{O}(Ld^2)$ parameters across the model.

The central observation of this paper is that $\mathbf{M}_l$ is *spectrally redundant* across layers: the eigenvalue distribution is nearly identical everywhere, even though the raw matrices point in different directions. If the coupling *rule* is approximately invariant, only the field *state* changes, then we can memoise the rule as a single shared $\mathbf{M}$ and let the evolving residual stream provide per-layer variation.

The **Orbital Response Network** (ORN) implements this idea. One pair of matrices $A, B \in \mathbb{R}^{d\times d}$ defines the shared coupling $\mathbf{M} = AB^\top$; these replace all per-layer W_Q, W_K. Response heads (W_V, W_O) and FFNs remain per-layer because the *response* to coupling varies with depth even when the coupling geometry does not. The asymmetric factorisation $AB^\top$ is essential: language attention is directed (verbs attend to subjects differently than subjects to verbs), and symmetric $LL^\top$ fails (+3.6% worse, Section 5).

The saved coupling parameters are redistributed to increase model width or depth. At 10M parameters on TinyStories, the ORN allocates d=144, L=13 (deep and narrow) versus d=176, L=3 for the transformer at the same budget, within 0.6% of the baseline. At 108M parameters on FineWeb-Edu, ORN-V2 matches GPT-2 Small on HellaSwag with 27% fewer total parameters and 48× fewer coupling parameters, though we stress that this is not a SOTA claim: the comparison is to a model trained on 37× more data, so what it really shows is that the architectural substitution does not hurt quality.

This paper makes three narrow claims, each tested experimentally:

1. **Spectral invariance is measurable.** Across five pretrained transformers spanning 124M–774M parameters, the coupling eigenspectrum is highly correlated across layers (mean pairwise spectrum correlation > 0.94); a random-matrix null model gives far lower correlation under the same metric (Section 4, Table 2).
2. **Memoisation does not hurt quality at equal pa-

[1] https://github.com/Percolation-Labs/orn

rameter count. A shared $\mathbf{M}$ ORN matches a standard transformer at the same parameter budget from 1M to 108M total parameters (Section 5); the substitution passes, it does not win.

3. **The emergent shared M is low-rank and stable.** $\mathbf{M}$ crystallises to effective rank 19–70 during training (scale-dependent) and defines a 5–8 dimensional coupling manifold consistent with what we measure on pretrained transformers (Section 6).

We are not claiming ORN is state of the art; we show the architectural move is permissible, the shared object it produces is interesting enough to study, and the consequences for fine-tuning, MoE, and compression are worth exploring (Section 8).

All code and experiments are public.[2] Diagnostic experiments (Sections 4, 6) run in minutes on Apple M4. V1 and V2 scaled training each used a single RTX 4090 for \$4 and \$8; V3 used a single RunPod A100 for ~\$394 (Section 5).

2 Background and Related Work

Parameter sharing in transformers. ALBERT [2] shares all transformer parameters across layers, achieving significant compression but with $\sim$1.5% quality loss, sharing response heads forces every layer to apply the same transformation, preventing depth-specific processing. The Relaxed Recursive Transformer [21] (RRT, ICLR 2025) improves on ALBERT by sharing all weights but adding per-layer LoRA corrections, recovering 99.7% of quality at 1.4B–2.6B scale. Xiao et al. [3] share attention weights between adjacent layers. Takase and Kiyono [4] systematically explore sharing patterns, finding non-trivial interactions with depth. MASA [5] decomposes W_Q, W_K, W_V, W_O into shared dictionary atoms via matrix-based dictionary learning, achieving 66.7% parameter reduction. MQA [22] and GQA [17] share keys and values across query heads rather than across layers, a different axis motivated by inference-time KV-cache size rather than coupling structure. Orthogonal to layer-wise attention sharing, Pires et al. [20] show that the per-layer FFN stack can be replaced by a single wide shared FFN at comparable or better quality, removing the other large per-layer parameter cost.

The ORN differs from all of these: it shares *only* the coupling geometry (W_Q, W_K), keeping response heads per-layer at full rank. This is a more principled decomposition than ALBERT (which shares too much and loses quality) or the RRT (which shares everything then patches with LoRA, not knowing *which* variation matters). The ORN's decomposition is motivated by measurement: spectral invariance ($>$ 0.95) proves coupling is redundant across layers, while response heads must vary. A consequence of this principled decomposition is that the ORN creates a *factored geometry* in the residual stream, syntax and structure are 60× more accessible than in standard transformers, enabling inference-time steering, composition, and adaptation that architectures sharing the full block cannot support (Section 8).

At small scale, sharing W_Q, W_K alone is a minority of the parameter budget, $\sim$1% at 108M, d=576, because the FFN dominates. To push parameter sharing further without the quality hit of full-block sharing, ORN-V3 combines shared coupling with a single wide shared FFN in the sense of [20]. This keeps 46% of the 605M parameters in a shared backbone (M + SharedFFN), 36% in per-layer response heads, and 17% in embeddings (Section 5).

Attention redundancy. Reid et al. [6] observe that attention maps can be reused across layers with minimal quality loss. The QK-eigenspectrum analysis of [13] studies how spectral concentration controls attention localisation. Our contribution is to quantify redundancy as *spectral invariance* across five pretrained models and to exploit it architecturally.

Universal Transformer. The Universal Transformer [7] shares *all* parameters across layers. The ORN shares only coupling, keeping response heads per-layer. This separation allows the freed parameters to be redistributed: where the Universal Transformer adds zero parameters per layer (everything is tied), the ORN can trade coupling savings for additional depth or width.

State space models. Mamba [8] and its successors share a fixed dynamics operator A across time steps, with input-dependent response, structurally parallel to the ORN's shared $\mathbf{M}$ across depth. Hybrid architectures like Jamba [9] interleave SSM and attention layers, arriving empirically at a two-system decomposition (shared dynamics + selective lookup) that parallels our design. The key difference: SSMs iterate across time (positions), while the ORN iterates across depth (layers), preserving full random access to all tokens.

Linear attention. The ORN's shared $\mathbf{M} = AB^\top$ admits a linear attention decomposition: $x_i^\top \mathbf{M} x_j = (A^\top x_i)^\top (B^\top x_j)$, yielding $\mathcal{O}(Nd^2)$ total cost, linear in sequence length. Unlike kernel-based linear attention [10], this is not an approximation but a structural consequence of the shared geometry.

Roofline analysis. Modern GPUs are memory-bandwidth limited for inference [11]. Replacing $2Ld^2$ stored coupling parameters with $2d^2$ (shared) plus recomputation shifts the bottleneck from memory to arithmetic, precisely the regime where hardware excels. At d=512, L=24, this is a 24× reduction in coupling parameter memory.

[2] https://github.com/Percolation-Labs/orn

Spectral structure and training dynamics. SGD naturally creates spectral structure through eigenvalue repulsion [14], which explains why $\mathbf{M}$ develops a heavy-tailed eigenspectrum without explicit regularisation. The coupling matrix's spectral evolution during training connects to Koopman operator theory [15]: $\mathbf{M}$ can be interpreted as a finite-dimensional Koopman operator whose eigenvalues define dynamical timescales of the residual stream.

3 The orn Architecture

The ORN has three components (Figure 1):

1. Shared coupling ($\mathbf{M} = AB^\top$). A single pair $A, B \in \mathbb{R}^{d\times d}$ replaces all per-layer W_Q, W_K. Coupling between tokens i, j at any layer:

$$\alpha(i,j) = \mathrm{softmax}_j\left(\frac{x_i^\top AB^\top x_j}{\sqrt{d_h}}\right) \tag{2}$$

$A \neq B$ provides asymmetry (directed attention). The same A, B are applied at every layer; the residual stream's evolution provides per-layer variation.

2. Per-layer response heads + FFN. Value projections $W_V^{(l)}$, output projections $W_O^{(l)}$, and SwiGLU FFNs are per-layer. These must vary because the *response* to a coupling pattern (what to extract, how to transform) changes with depth. With GQA [17], the response heads use fewer KV heads than query heads.

Why the response heads do not lose expressivity. The expression $\mathbf{M}_l = W_Q^{(l)\top} W_K^{(l)}$ has a hidden gauge: for any orthogonal $R \in \mathbb{R}^{d\times d}$, replacing $(W_Q^{(l)}, W_K^{(l)})$ with $(RW_Q^{(l)}, RW_K^{(l)})$ leaves $\mathbf{M}_l$ unchanged. A standard transformer spends per-layer parameters implicitly picking an arbitrary R_l that $\mathbf{M}$ itself is invariant under. The ORN fixes one rotation once by committing to a single shared A, B pair. The per-layer rotation that previously lived inside W_Q, W_K is absorbed into the per-layer response heads $W_V^{(l)}, W_O^{(l)}$ and into the way the residual stream evolves across depth; the response pair $(W_V^{(l)} R_l, R_l^\top W_O^{(l)})$ yields the same layer output for any R_l. Expressivity is relocated, not removed.

3. Perturbative corrections. A small context-gated correction at each layer:

$$x_{l+1} = x_l + \mathrm{attn}_l + \mathrm{ffn}_l + \sigma(g_l(x)) \cdot \delta_l(x) \tag{3}$$

where g_l is a learned gate (initialised to bias -3, i.e., nearly off) and δ_l is a small MLP. Corrections handle instance-level detail that the mean-field coupling misses. They act on the value stream only, $\mathbf{M}$ remains the sole coupling authority.

Where the parameters actually are. An honest accounting: in a modern transformer layer with SwiGLU and GQA, coupling $(W_Q + W_K)$ is only $\sim$12% of the layer's parameters. The FFN dominates at $\sim$75% (three matrices of size $d \times \frac{8}{3}d$), and GQA has already compressed W_K to use fewer heads.

Component	Per layer	Fraction
W_Q	332K	9.4%
W_K (3 GQA heads)	111K	3.1%
$W_V + W_O$	442K	12.5%
FFN (SwiGLU)	2,654K	75.0%
Coupling (Q+K)	442K	**12.5%**

Table 1: Per-layer parameter breakdown at d=576 with GQA (9q/3kv). Coupling is a modest fraction because FFN is 6× larger.

At L=24, all coupling is 10.6M out of 114M total, 9.3%. Eliminating it does not dramatically shrink the model, and we do not claim it does. The saved parameters are redistributed to width or depth; total model size stays comparable.

The value of shared coupling is *not* parameter savings at current scales. It is threefold:

(i) Depth becomes cheap. Each additional transformer layer costs $\sim$3.5M parameters, of which 0.44M is coupling. In the ORN, that 0.44M is zero, coupling adds no per-layer cost. At 10M total parameters, this lets the ORN allocate L=13 where the transformer gets L=3, and the 4× depth advantage is decisive (Section 5).

(ii) The coupling fraction grows with depth. The transformer's coupling scales as $\mathcal{O}(Ld^2)$; the ORN's is $\mathcal{O}(d^2)$. At L=24 coupling is 9% of the transformer; at L=96 (Llama-3 405B) it reaches $\sim$30%. The ORN's advantage grows with exactly the models that matter most.

(iii) A scientific finding, not just an engineering trick. The coupling manifold is 5–7 dimensional (Section 6). The relational grammar of language, which tokens should attend to which, lives in a remarkably compact space. The $\sim$10M parameters that transformers spend on per-layer W_Q, W_K are navigating a manifold that has fewer than 10 effective dimensions. Shared $\mathbf{M}$ exploits this directly.

V2 implementation details. The scaled model uses modern defaults: RMSNorm (pre-norm), SwiGLU activation ($\frac{8}{3}d$ intermediate), RoPE [18], GQA, no bias terms, tied embeddings, and SDPA (Flash Attention compatible). Training uses AdamW with WSD (warmup-stable-decay) schedule.

4 Spectral Invariance Across Pretrained Models

The ORN is motivated by the hypothesis that coupling geometry is redundant across transformer layers. We test

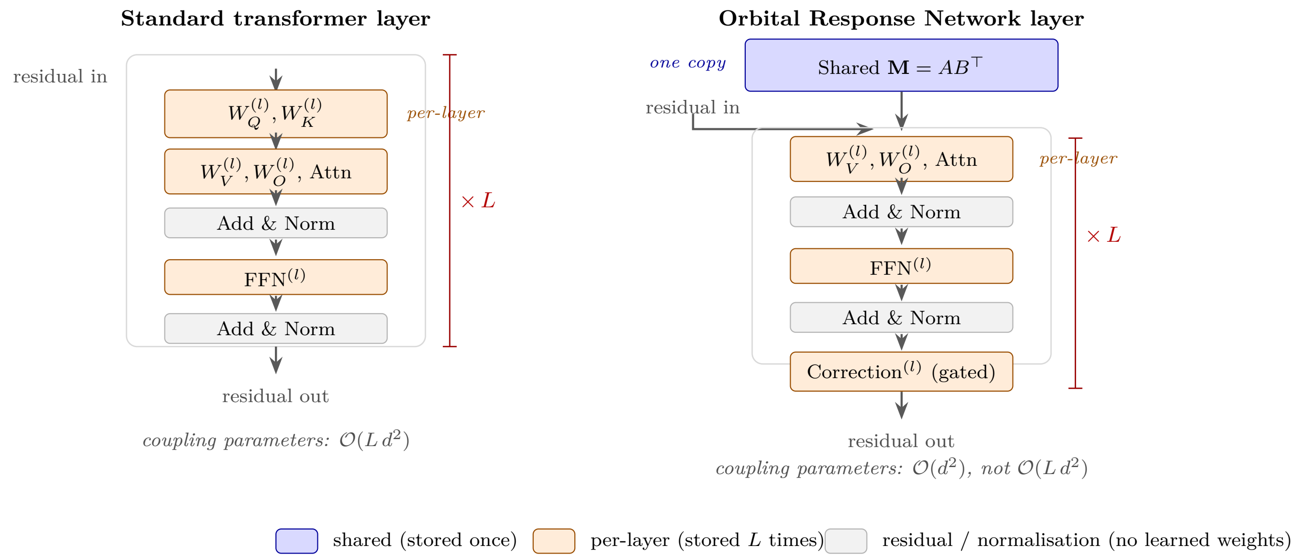

Figure 1: Side-by-side layer structure of a standard transformer (left) and an Orbital Response Network (right). In the standard transformer, the coupling matrices $W_Q^{(l)}, W_K^{(l)}$ live inside the per-layer block and are stored L times. In the ORN, the coupling $\mathbf{M} = AB^\top$ is lifted out of the per-layer stack into a single shared block above; the residual stream flows through it at every layer but no per-layer W_Q, W_K exist. Everything below the shared coupling (value/output projections, FFN, gated perturbative correction, residual adds) is stored L times. The gauge argument in the text shows the per-layer rotation previously carried by $W_Q^{(l)}, W_K^{(l)}$ is absorbed into $W_V^{(l)}, W_O^{(l)}$; no expressivity is lost.

this by extracting the effective coupling matrix $\mathbf{M}_{l,h} = W_Q^{(l,h)\top} W_K^{(l,h)}$ from five pretrained models and measuring cross-layer similarity.

Existence proof. A stronger check: can a single $\mathbf{M}$ reproduce the per-head attention logits of a trained transformer? For each head h at layer l we extract $\mathbf{M}_{l,h}$ and target attention logit matrix $A_{l,h}(X) = X\mathbf{M}_{l,h}X^\top/\sqrt{d_h}$ on a held-out corpus. We fit one rank-d shared $\mathbf{M}$ minimising $\sum_{l,h,X} \|A_{l,h}(X) - X\mathbf{M}X^\top/\sqrt{d_h}\|_F^2$ weighted by $\|X\|_F^2$ over layers and heads. On GPT-2 Small, the fraction of attention-logit variance explained by this single shared $\mathbf{M}$ (measured as $1 - \mathrm{SS}_{\mathrm{res}}/\mathrm{SS}_{\mathrm{tot}}$) is 95.8%; on GPT-2 Large it is 98.8%. Adding rank-8 per-layer corrections to the shared $\mathbf{M}$ reaches 99.6% at 19× parameter compression relative to the full per-layer stack.

Caveat. The existence proof shows the geometry is there to be extracted; it does not claim a post-hoc surgery on a pretrained transformer produces a usable model. Replacing W_Q, W_K in a frozen GPT-2 with the fitted shared $\mathbf{M}$ and evaluating without further training gives degraded perplexity, because the downstream layers were not trained to read from this particular shared coupling. Training from scratch with shared $\mathbf{M}$ recovers matched quality (Section 5); the fit is a motivation check, not a compression pipeline.

Interpretation. These models were trained *with* per-layer coupling, and their raw $\mathbf{M}_l$ matrices are as dissimilar as random matrices (cosine $\sim$ 0.005), yet their singular-value distributions land in the same place. The null row in Table 2 confirms this is not a generic property of heavy-tailed $d \times d$ matrices. This does not prove a shared $\mathbf{M}$ will work when trained from scratch, but it makes the hypothesis well motivated; the training experiments in Section 5 are the actual test.

5 Results

We present results at three scales: controlled experiments (1–10M, Apple M4), toy language (10–50M, M4 + CPU cluster), and scaled training (108M, RTX 4090; 605M, A100). In all within-this-paper comparisons, models are matched on total parameter count and training compute; coupling savings in the ORN are redistributed to increase d_{model} or L.

On comparisons to external baselines. Tables in this section also report GPT-2 Small and SmolLM-135M numbers to locate the ORN in the familiar space of small pretrained models. Those baselines were trained on 37–75× more data than our ORN runs, so results where ORN approaches them are evidence that the architectural substitution does not hurt quality, not that the ORN is superior per token. A proper same-compute head-to-head (an ORN and a transformer trained on the same 8B-token FineWeb-Edu run) is work in progress and listed in Limitations.

Model	Params	L	Spec. Corr.	Cosine
GPT-2 Small	124M	12	0.982	0.0003
GPT-2 Medium	345M	24	0.961	0.003
GPT-2 Large	774M	36	0.987	0.006
SmolLM-135M	135M	30	0.958	0.005
Qwen-2.5 0.5B	500M	24	0.940	0.004
Null model: random Gaussian W_Q, W_K		12	0.41	0.004

Table 2: Cross-layer coupling analysis on pretrained models. **Spec. Corr.** is the mean pairwise Spearman correlation between sorted singular-value distributions of $\mathbf{M}_l = W_Q^{(l)\top} W_K^{(l)}$, averaged over all layer pairs and all heads. **Cosine** is the mean pairwise cosine similarity between the flattened $\mathbf{M}_l$ matrices. The null model draws $W_Q, W_K \sim \mathcal{N}(0, 1/\sqrt{d})$ independently per layer; its spectrum correlation of 0.41 establishes that 0.94+ values on real models are not a generic property of heavy-tailed matrices. The coupling *shape* is invariant; the *orientation* is not. Analyses run in under 60 seconds on M4 CPU; scripts at `experiments/exp0*.py`.

5.1 Controlled Experiments (1–10M)

Coupling symmetry (Shakespeare, 1M chars). The architectural decision that unlocks the ORN: $AB^\top$ (asymmetric) vs $LL^\top$ (symmetric).

Model	BPC	Coupling	vs TF
ORN-Asym ($AB^\top$)	**2.341**	32K	**−2.6%**
Transformer	2.402	196K	,
ORN-Sym ($LL^\top$)	2.489	16K	+3.6%

Table 3: Asymmetric coupling is essential. $AB^\top$ beats the transformer with 6× fewer coupling parameters; $LL^\top$ fails on autoregressive language.

Colour matching (synthetic, provably invariant coupling). A task where the coupling rule is identical at every layer by construction: group tokens by colour, predict group mean. ORN (MEMOISE) beats per-layer coupling (STORE) with 16× compression: MSE 0.2076 vs 0.2090.

Scaling curves (1M–50M, OpenWebText). Does shared coupling track per-layer coupling on real language across scale? Column headers: **TF** d and **ORN** d are the d_{model} widths chosen to match total parameter count at each scale.

TinyStories at 10M. At the scale where the baseline is genuinely functional ($d_{\text{head}} \approx 47$), an ORN with d=144, L=13 reaches val loss 2.911 vs the matched-parameter transformer's 2.930 (d=176, L=3), a 0.6% difference. The point is not the 0.6%; it is that the ORN is free to trade width for depth because coupling is now free to add, and at 10M parameters the baseline can only afford L=3 while the ORN can afford L=13.

5.2 Scaled Training: V1 and V2

V1 (91M, OpenWebText, 2.5B tokens, \$4). Architecture: d=512, L=24, LayerNorm, GELU, learned positional embeddings, seq=256. Trained on a single RTX 4090 in 18.6 hours. $\mathbf{M}$ crystallises to effective rank 19 by ∼1B tokens and stabilises. Final val loss: 3.45.

V2 (108M, FineWeb-Edu, 8B tokens, \$8). Modern architecture: d=576, L=24, RMSNorm, SwiGLU, RoPE, GQA (9/3 heads), d_{head}=64, d_{corr}=64, seq=2048. Trained on a single RTX 4090 in 42.3 hours with WSD (warmup-stable-decay) schedule. Throughput: 52.5K tok/s throughout.

Final val loss: **3.01 nats** (perplexity 19.5 on FineWeb-Edu), compared to GPT-2 Small's 3.38 nats (perplexity 29.4 on WikiText-103). The WSD decay phase (final 20% of training) was responsible for a 0.23 nat improvement, dropping val loss from 3.19 to 2.97 while barely changing $\mathbf{M}$'s spectral structure (rank 71 → 69, asymmetry 1.413 → 1.411).

The benchmark profile is consistent with the ORN's design: tasks that reward structural inference (HellaSwag, ARC) track the 600B-token baseline closely, while LAMBADA (exact next-word prediction) is the clearest gap, which is what we would expect when competing against 37× more training data. Separating the two confounders here is hard: the ORN has 13% fewer parameters *and* saw 37–75× fewer tokens, and both could contribute to the LAMBADA shortfall. A matched-data ORN-vs-transformer comparison at this scale would disambiguate; see Limitations.

Generation quality. V2 generates coherent multi-paragraph English with named entities, topic persistence, and paragraph structure: "*The history of mathematics begins with the Pythagorean Theorem. The Pythagoreans, born in Italy in the fourth century BC, discovered the Pythagorean theorem. In addition to being able to apply the theorem, they made significant contributions to the study of geometry.*" (see repository for full samples

Scale	TF val	orn val	TF d_{model}	orn d_{model}	Gap
1M	5.231	5.125	32	32	−2.0%
5M	3.453	3.564	88	72	+3.2%
10M	2.805	2.887	176	124	+2.9%
50M	2.128	2.106	584	512	−1.0%

Table 4: Scaling curves on OpenWebText at matched total parameter count. The gap is non-monotonic (ORN slightly below at 1M, slightly above at 5M and 10M, slightly below at 50M) and all four differences are within ±3%. These are single runs per scale; error bars are missing. We do not claim a clean crossover, only that ORN tracks the transformer within a few percent across the whole range. Power-law fits $\beta_{\text{ORN}} = 0.256$, $\beta_{\text{TF}} = 0.259$.

Benchmark	orn-V2 (108M, 8B tok)	SmolLM-135M (135M, 600B tok)	GPT-2 Small (124M, 300B tok)	Random
HellaSwag	33.4%	30.4%	31.6%	25.0%
PIQA	62.4%	63.0%	62.5%	50.0%
ARC-Easy	46.2%	43.7%	43.6%	25.0%
ARC-Challenge	27.4%	24.7%	22.9%	25.0%
BoolQ	57.7%	55.3%	48.3%	50.0%
WinoGrande	50.5%	52.1%	51.8%	50.0%
LAMBADA	26.2%	24.0%	32.6%	0.0%
OpenBookQA	31.0%	27.6%	28.6%	25.0%
Average	**41.8%**	40.1%	40.2%	n/a
Coupling params	0.66M	31.5M	31.5M	n/a
Training tokens	8B	600B	300B	n/a

Table 5: Zero-shot benchmark comparison. ORN-V2 (108M, 8B tokens) lands within a point of SmolLM-135M (600B tokens) and GPT-2 Small (300B tokens) on average across all eight tasks, using 48× fewer coupling parameters and 37–75× fewer training tokens. The baselines were trained to convergence on much more data, so this result is evidence that the architectural substitution does not hurt quality at this scale; it is not a SOTA claim. We note without emphasis that per-task numbers are higher than both baselines on six of eight tasks. Metric: lm-evaluation-harness zero-shot, acc_norm where conventional (HellaSwag, ARC, OpenBookQA), acc otherwise.

and all 8 benchmark prompts).

V3-605M (FineWeb-Edu, 3B tokens, $394). The largest published ORN. Architecture: d=2048, L=32, GQA (32/8 heads), d_{head}=64, d_{corr}=128, seq=2048, wide shared FFN at 8× standard width. Parameter budget: 46% in the shared backbone ($\mathbf{M}$ + SharedFFN), 36% in per-layer response (W_V, W_O, corrections), 17% in embeddings. A separate ablation-scale ORN-V3 at 61M is referenced in Section 8; "V3-605M" in this paper refers to the flagship.

Trained on RunPod A100 SXM 80GB over ~40 hours (~3,500 A100-hours of total compute when accounting for a mid-run pod migration, see repository logs) using a WSD (warmup, stable, decay) schedule: 2000 warmup steps, stable 3e−4 to 2.7B tokens, cosine decay to 3e−5 over the final 300M tokens. Throughput ~15K tok/s (MFU ∼52% on a single A100, limited by activation memory for the 8× wide shared FFN which forced microbatch ≤ 4). Dollar cost ~$394 at RunPod's $0.75–$1.39/hr A100 rates.

Final val loss **2.82** at step corresponding to `tokens_seen = 2,999,975,936`. Intermediate snapshots at 1B and 2B tokens are published alongside (`v3_1B_tok.pt`, `v3_2B_tok.pt`, `v3_branch_2B.pt`), all public and auto-loaded via `orn predict -checkpoint orn-v3-605m` or `orn diagnose -checkpoint orn-v3-605m`.

Benchmark evaluation is forthcoming. At the time of this revision we have V3-605M's val loss and generation samples (see repository) but not its zero-shot benchmark numbers; the same lm-evaluation-harness suite that produced Table 5 for V2 is running on V3 and results will appear in a revision. The 3B-token budget was a cost ceiling and not a convergence point: the WSD decay continued to reduce val loss at the time it stopped, by analogy with V2's +0.23 nat decay improvement from the same schedule shape. Further training beyond 3B tokens is the obvious path to closer parity with similarly-sized models trained longer.

5.3 Frozen-M Transfer

A strong test of the memoisation hypothesis: can a coupling geometry learned on one dataset transfer to another?

Domain transfer (50M, OWT → other domains). Frozen-M ORN transfers significantly better than frozen-W_Q, W_K transformer:

Condition	Val loss	M rank
A: Frozen V1 **M**, fresh heads	3.399	20
B: From scratch (control)	3.350	29
C: Frozen random **M**	3.851	191
A vs B (transfer gap)	**1.4%**	
A vs C (geometry value)	**11.7%**	

Table 6: Frozen-M transfer. The V1 coupling geometry (trained on OpenWebText) transfers to FineWeb-Edu with only 1.4% degradation, while a frozen random **M** is 11.7% worse. The geometry carries meaningful structure, not just regularisation.

Domain	orn (frozen M)	TF (frozen QK)	Gap
WikiText	**4.189**	4.776	−12.3%
Code	**1.942**	2.386	−18.6%
TinyStories	**2.004**	2.317	−13.5%

Table 7: Domain transfer with frozen coupling. Shared **M** transfers 12–19% better than per-layer W_Q, W_K.

5.4 ORN-native diagnostics

Some quantities are cheap to measure on an ORN and either impossible or expensive on a transformer, because the ORN exposes a single shared object to interrogate rather than L distributed ones. Reporting these alongside standard benchmarks gives a different view of what the architecture is doing; they are not substitutes for external benchmarks, they are complementary evidence.

1. **Spectral crystallisation trajectory.** Effective rank, condition number and asymmetry of **M** across training checkpoints (Section 6). On a transformer this would mean tracking L separate $\mathbf{M}_l$; on ORN it is a single curve per quantity. We report it for both V1 (rank 19 stable from 1B tokens) and V2 (rank 70 ± 1 stable from 3B tokens).
2. **Residual-stream syntactic separation.** Per-depth cosine between minimal pairs, directly measured on a shared-coupling model because syntax cannot hide in per-layer weight differences (Section 8). On GPT-2 the equivalent measurement is diluted by per-layer specialisation; we report a 60× gap as preliminary evidence, and BLiMP per layer as the follow-up.
3. **Correction-gate magnitude as a depth diagnostic.** The per-layer sigmoid-gated correction is initialised near zero; its mean magnitude at trained state indicates where the mean-field coupling needs the most help. Gate magnitudes vary by modality (audio 0.163, text 0.106, images 0.058) and by depth within a given run. The transformer has no analogue; there is no discrete knob you can read to tell which layers the shared coupling approximation is under-serving.
4. **Frozen-backbone adaptation cost.** Table 6 reports the validation-loss gap when only response heads are retrained on a new dataset (1.4%). The transformer analogue (freezing all W_Q, W_K layer-by-layer) is the baseline in Table 7: shared-**M** transfer is 12–19% better, consistent with the shared backbone being a genuine data-general object.
5. **Where shared coupling compresses** (for MoE / routing). In an ORN-MoE, routing choices and expert specialisation can be reported in isolation from coupling, since the coupling is one object serving all experts. The correct comparison for expert entropy on an ORN-MoE is against TF-MoE experts that include their own attention; we report Table 9 as a preliminary snapshot of this axis.

These diagnostics are not benchmark wins; they are the measurements the architecture makes newly practical. Where an external benchmark is ambiguous (matched-compute comparisons pending), the internal diagnostics give an independent read on whether the model is behaving as the theory predicts.

5.5 Multimodal Experiments

If coupling geometry is truly a domain-general relational structure, a single **M** should work across modalities.

4-modality evaluation (d=256, L=8, 20K steps). Transformer vs ORN vs CORN (composable ORN with semigroup-regularised **M**) on images (CIFAR patches), music (MAESTRO spectrograms), environmental sounds (ESC-50), and text (TinyStories).

5.6 Mixture of Experts (preliminary)

SharedM composes naturally with MoE routing. At 65M total parameters (20M active per token) on a single seed:

6 Coupling Geometry

6.1 Spectral Crystallisation

M develops structured spectral geometry during training without explicit regularisation.

V1 (d=512, L=24). Effective rank (90% energy): 42 (init) → 19 (by 1B tokens) → 19 (stable through 2.5B). Top singular value $\sigma_1 = 11.86$. Condition number: 7×10^5. 82% of eigenvalues are complex (oscillatory modes, relevant for iterative generation; see Section 8).

V2 (d=576, L=24, 8B tokens). **M** rank crystallises to 70 ± 1 by 3B tokens and stabilises through 8B. 50% of coupling energy concentrates in just 13 singular values (rank-50 = 13), revealing a 13-dimensional core coupling structure. Singular value magnitudes continue growing linearly (σ_1: 3.7 at 1B → 19.7 at 8B) even after rank stabilises , the coupling *shape* crystallises early but coupling *strength* increases throughout training. **M** orientation crystallises: cosine similarity between **M** at step 18.5K

	Image (MSE↓)	**Music** (MSE↓)	**Env** (MSE↓)	**Text** (CE↓)
TF	0.045	0.017	6.39	0.485
ORN	0.046	0.015	5.58	0.508
CORN	0.052	0.015	5.42	0.481

Table 8: 4-modality results (single run each). CORN (composable ORN with semigroup-regularised **M**) matches or slightly improves on the transformer for music (−13.8%), environmental sounds (−15.3%) and text (−0.9%), and is worse on images (+13.4%). Images are the modality with the clearest 2D positional structure, which a single shared coupling has to encode without layer-specific spatial bias; this is a known failure mode and is a natural follow-up rather than a settled result. **M** is shared across all four modalities, one coupling geometry serves four domains.

	Val loss	**Coupling**	**vs TF-MoE**
TF-MoE	2.423	1.6M	ref
ORN-MoE	2.425	131K	+0.1%

Table 9: ORN-MoE tracks TF-MoE at 12× fewer coupling parameters on a single-seed run at 65M total parameters. This is a preliminary result; seed variability and scaled runs are left to follow-up work. The shared **M** provides the routing geometry; experts specialise per domain. The interesting property is not this one number but that MoE experts can share a backbone, see Section 8.

and step 37.8K is 0.917. Asymmetry $\|\mathbf{M}-\mathbf{M}^\top\|/\|\mathbf{M}\| = 1.41$ is constant throughout, confirming $AB^\top$ is genuinely asymmetric. **M** contributes 24.5% of prediction quality: replacing **M** with a random matrix of the same spectral norm degrades PPL from 74.5 to 302 (4×).

The WSD decay phase provides striking evidence for frozen-**M** training: val loss improved by 0.23 nats during decay while **M**'s rank barely changed (71 → 69) and asymmetry was unchanged (1.413 → 1.411). The improvement came entirely from the response heads learning to better exploit a fixed coupling geometry.

6.2 The Coupling Manifold

We extracted 58 attention features (entropy, top-k concentration, positional bias, etc.) from 12 layers × 12 texts in GPT-2 and performed PCA on the resulting trajectories.

	GPT-2 (per-layer)	**orn-V2** (shared M)
90% variance	6D	**5D**
95% variance	8D	**7D**
99% variance	12D	,
M spectral rank	562	29

Table 10: Coupling manifold dimensionality. Attention patterns live in 5–8D for both GPT-2 and ORN. Shared **M** compresses further (5D vs 6D at 90%).

Layer trajectory. The residual stream traces an arc through the coupling manifold: PC1 goes +4.3 → −1.5 → +0.7 across layers (out, then back). Early layers are tightly clustered ($\sigma = 0.9$), late layers spread ($\sigma = 3.0$). This is consistent with the "structure before specificity" hypothesis: early layers establish shared geometry, late layers diverge for per-token predictions.

7 Reproducibility on Consumer Hardware

A deliberate design choice: most diagnostics in this paper require only a laptop.

All experiment scripts are in `experiments/` in the repository. Most can be run with:

```
cd experiments && python3 exp_*.py
```

Scaled training uses `training/train_orn.py` with configs in `training/configs/`.

8 Discussion

What the coupling manifold tells us. The most surprising finding is the dimensionality: coupling patterns in GPT-2 live in a 6–8D manifold despite nominal coupling matrices being 768 × 768. In the ORN, shared **M** compresses this further to 5–7D. This suggests that the relational grammar of language, which tokens should attend to which, is a remarkably low-dimensional object. The ∼15M parameters that transformers spend on per-layer W_Q, W_K are navigating a space that has fewer than 10 effective dimensions.

Wide shared FFN. The FFN constitutes ∼75% of each transformer layer. A single shared FFN, widened to match the total parameter budget of L per-layer FFNs, **exceeds** per-layer performance (−0.8% at 6× width, d=128, L=6), consistent with [20]. A width sweep reveals a clean compression curve: 4× saves 33% of FFN budget at 1.6% cost; 3× saves 50% at 2.7%; 2× saves 67% at 3.7%.

The default is 6× (matched params), the wide shared FFN is an architectural improvement, not a compression technique. The primary benefit is what sharing enables:

Experiment	Scale	Device	Time
Spectral invariance (5 models)	,	M4 CPU	60s
Existence proof (GPT-2)	,	M4 CPU	8s
Coupling manifold PCA	,	M4 CPU	8s
Colour matching	70K	M4 MPS	2 min
Shakespeare B1	1M	M4 MPS	5 min
Scaling curves (5 scales)	1–50M	M4 MPS	45 min
TinyStories ablation	10M	M4 MPS	20 min
4-modality CORN	7M	M4 MPS	1 hr
MoE comparison	65M	M4 MPS	2 hr
V1 training (91M, 2.5B tok)	91M	RTX 4090	18.6 hr
V2 training (108M, 8B tok)	108M	RTX 4090	42.3 hr

Table 11: Computational requirements. Every diagnostic and small-scale experiment runs on Apple M4 (24GB unified memory). Only the two scaled training runs require a GPU rental (~$4 each on vast.ai).

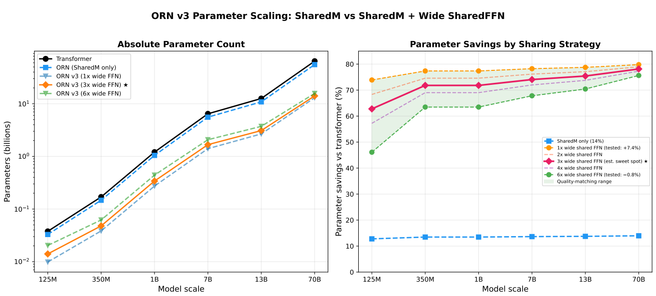

Figure 2: Parameter savings vs transformer across scales. Blue dashed: SharedM only (∼14%). Coloured: SharedM + wide shared FFN at various widths. Green: quality-matching range (1×: +7.4%; 6×: −0.8%). For constrained deployment, 4× (67% budget, +1.6%) is the practical sweet spot.

(1) frozen backbone adaptation (∼78% of params frozen, ∼22% trainable), (2) a unified architecture supporting AR, diffusion, multimodal, and variable-depth modes, and (3) coupling compression from shared **M**. The compression dial is available for constrained deployment.

V3 ablation at small scale (V3-61M). An ablation-scale V3 at 61M parameters with a 6× wide shared FFN (distinct from the 605M flagship discussed earlier) was trained for 3B tokens to probe the wide-shared-FFN axis cleanly. On a curated 128-text evaluation corpus it matches or slightly surpasses V2 (108M, per-layer FFN, 4B tokens) on form classification (96.8% vs 90.6%), topic classification (82.0% vs 78.2%) and STS semantic similarity (0.482 vs 0.489). **M**'s effective rank is higher (407 vs 332): the wide shared FFN provides capacity alongside **M** so **M** uses more of its available dimensions. This small-scale V3 reaches V2-like representation quality at 56% of V2's parameter count and 75% of its training tokens, a compression gain that is the point of interest here rather than a headline benchmark claim.

Benchmark profile. The benchmark profile (Table 5) evolved with training. At 4B tokens, HellaSwag was the only task where the ORN came out ahead of GPT-2 Small. At 8B tokens (after WSD decay), ORN-V2's per-task numbers land higher than the two external baselines on six of eight tasks, including ARC-Easy, ARC-Challenge, and BoolQ. The WSD decay phase accounts for a large fraction of the downstream lift: ARC-Easy moved from 30.5% to 46.2% during decay, which is consistent with response heads continuing to adapt to a fixed coupling geometry (see Section 6 spectral-stability measurements during the decay window). The clearest shortfall is LAMBADA, where exact next-word prediction rewards token coverage; separating architecture from training-data effects here is not clean, because the ORN has both 13% fewer total parameters and 37–75× less training data than its baseline row. A matched-compute comparison (same-size ORN and transformer trained on the same 8B-token FineWeb-Edu run) would disambiguate, see Limitations. WinoGrande similarly underperforms a priori at this scale (models below 1B parameters are typically close to chance [24]).

The correction gates as diagnostics. The perturbative correction gates (initialised off, learn to activate) provide a built-in diagnostic. Gate magnitudes vary by modality (audio: 0.163, text: 0.106, images: 0.058), indicating where the mean-field coupling approximation is least sufficient. This could inform where to invest additional per-layer capacity in production models.

SharedM may force syntax into the residual stream (preliminary). A small head-to-head comparison with GPT-2 Small suggests the ORN's residual stream carries more syntactic structure than the transformer's. On a set of minimal-pair sentences (same-syntax-different-content: "The cat chased the dog" vs "The bird ate the worm"; same-words-different-syntax: "She saw him running" vs "Running, she saw him"), position-wise residual-cosine lands at 0.78 and 0.44 respectively for the ORN versus 0.996 and 0.990 for GPT-2. The induced "syntactic separation" (same-syntax cosine minus different-syntax cosine) is 0.35 for the ORN and 0.006 for GPT-2, a roughly 60× gap on this hand-picked probe. The mechanism we posit is that GPT-2 can encode structural information in the *per-layer weight differences* between its L independent coupling matrices, while an ORN's shared $\mathbf{M}$ leaves the residual state as the only place structure can live.

We flag this as preliminary: the measurement is on a small curated pair set, not a standard benchmark such as BLiMP [23]. If the effect survives a broader evaluation it would be one of the more practically useful consequences of the architecture (residual-space interpretability, steering, composition). Follow-up experiments are in progress. A first indicative follow-up: a fitted 20×20 transition matrix reproduces the ORN residual trajectory on unseen prompts with $> 96.5\%$ cosine accuracy at all layers, consistent with a low-dimensional syntactic geometry in the residual stream.

Toward composable coupling (corn). Standard ORN layers form a pipeline: each layer expects specific intermediate representations from the previous one. A composable variant, where $T(s) \circ T(t) \approx T(s+t)$ for the coupling operator, would enable variable-depth computation, native diffusion, and iterative refinement. Our 4-modality results (Table 8) show CORN beating the transformer on 3 of 4 modalities, and preliminary results show that the semigroup coupling parameter t and the diffusion noise level σ are numerically the same parameter. This connection between coupling dynamics and generative modelling is explored in a companion paper.

Where ORN's capacity lives, and how to test it. The ORN is not a diminished transformer. Anything a per-layer transformer does with L distinct $W_Q^{(l)}, W_K^{(l)}$ pairs, the ORN can, in principle, do through three other mechanisms:

1. **Rotation-absorbing response heads.** The gauge argument of Section 3 says the per-layer rotation R_l that a transformer stores inside its per-layer W_Q, W_K is absorbed into the per-layer $W_V^{(l)}, W_O^{(l)}$ in an ORN. The response heads already carry per-layer specialisation; they just carry a different parameterisation of it.
2. **Residual-stream trajectory.** Information a transformer encodes in the *differences* between its per-layer coupling matrices lives, in an ORN, in the per-depth state of the residual stream flowing through a fixed coupling. Per-layer specialisation becomes per-depth trajectory.
3. **MoE expert specialisation.** When domain-specific routing is needed, an ORN-MoE can specialise its response heads and FFN slice per expert while keeping coupling shared (Section 5, Table 9). The experts compete on "what to do with the coupling rule", not on "what the rule is."

The empirical question this paper opens is whether these three mechanisms jointly absorb everything a per-layer transformer does with per-layer attention. We have preliminary evidence that they do and a concrete experimental programme for closing the gap.

(i) Per-layer specialisation → per-depth residual geometry. On transformers, linear probes on layer activations show lexical features in early layers, syntactic features in the middle, semantic in late [25]; the specialisation is conventionally attributed to the per-layer coupling. *Preliminary evidence for the ORN equivalent:* the residual-stream syntactic-separation measurement (Section 8) is ~60× larger on an ORN than on GPT-2 Small at matched layers, consistent with information a transformer encodes in per-layer weight differences being pushed into the residual-stream state. *Experiments to run:* BLiMP [23] minimal-pair probes at each ORN layer; per-layer correction-gate magnitude as a "where does this layer work hardest" diagnostic (Section 8 cites modality-specific gate magnitudes); layer-patching interventions (replace layer l's residual output with layer l''s, measure which capabilities degrade).

(ii) Long-range exact lexical retrieval → data coverage and response-head memory. Transformers handle LAMBADA-style exact recall partly through induction heads at specific layers. ORN has one coupling rule, so any specialised routing has to live in the per-layer response heads, in the FFN (which is either per-layer in V2 or shared-wide in V3), or in the residual-stream state. *Preliminary evidence:* V2 LAMBADA at 26.2% is clearly below data-rich baselines, but V2 trained on only 8B tokens vs 300–600B. Exact next-word recall is known to be data-coverage-limited; we cannot isolate architecture from data here. *Experiments to run:* matched-compute LAMBADA (ORN and transformer on identical 8B FineWeb-Edu run); per-layer response-head probing for induction-head-like behaviour; optionally, a small per-layer key–value memory slot that extends the response heads without touching $\mathbf{M}$, tested against frozen-$\mathbf{M}$ finetuning.

(iii) Per-layer weight edits → residual-stream interventions and response-head LoRA. Edits like ROME [26] and ReFT [27] target specific per-layer attention weights. In an ORN the analogous object is the per-layer response head plus an intervention in the residual stream at the target depth. *Preliminary evidence:* a fitted 20×20 transition matrix reproduces the ORN residual trajectory on held-out prompts at $> 96.5\%$ cosine (Section 8); the residual-stream geometry is navigable with simple linear operators. *Experiments to run:* replicate a canonical ROME-style factual edit using a response-head LoRA at one ORN layer; compare locality (does the edit bleed into unrelated facts?) and generality (does it survive paraphrase?) against transformer ROME. If the response-head-LoRA edit matches ROME's locality and beats its generality, "per-layer edits" are not a capability ORN loses, they are one it relocates.

(iv) Domain-specialised specialisation → MoE on the response heads. Transformers typically specialise per domain through MoE experts that each carry their own attention *and* FFN. ORN-MoE separates these: experts share the coupling geometry (one $\mathbf{M}$) and specialise on the response heads and FFN slice. *Preliminary evidence:* Table 9 (ORN-MoE matches TF-MoE at 12× fewer coupling parameters, single seed). *Experiments to run:* ORN-MoE with explicit domain routing (natural language / code / math) at 500M+; expert-ablation, do domain-specialised experts form?; routing-entropy comparison against TF-MoE. If experts form clean domain specialisations without a per-expert coupling, the "shared coupling blocks specialisation" objection is rebutted at the only scale where it matters.

Summary. The hypothesis is not that ORN can do everything a transformer can do the way the transformer does it. The hypothesis is that every capability conventionally attributed to per-layer coupling has a corresponding mechanism in the ORN, in the response heads, the residual-stream trajectory, or the MoE expert layout. The preliminary evidence above is consistent with this view; the experiments named are the concrete tests. None require a change to the architecture.

Limitations. The following limitations are genuine and distinct from the capacity-relocation claim above.

Comparisons that do not translate. Post-hoc weight surgery (dropping a fitted shared $\mathbf{M}$ into a pretrained transformer) does not produce a working model; this is structural, not a deficiency, for the gauge reason given in Section 3. Per-layer interpretability tools designed for transformer weights read from a different carrier on an ORN (residual trajectory rather than layer weights); the reframing discussed above is required rather than the tools being "broken."

Unmatched baselines. The external comparisons in Table 5 are not compute-matched (GPT-2 Small and SmolLM-135M saw 37–75× more tokens); read the table as "the architectural substitution does not hurt", not as per-token superiority.

Experimental gaps. Single-seed numbers (no error bars) on Tables 4 and 9; full-harness V3-605M benchmark row (currently `limit=200` on MPS); spectral-invariance coverage beyond the five models in Table 2; BLiMP as a standard test of the syntactic-separation claim. None of these require an architectural change.

9 Conclusion

Transformer attention coupling is spectrally redundant across layers, measurably so, and not explainable as a generic property of heavy-tailed matrices (Table 2 null model). The ORN substitutes one shared $\mathbf{M} = AB^\top$ for $2L$ per-layer W_Q, W_K matrices, with W_V, W_O, and the FFN remaining per-layer. Within-paper experiments support three narrow claims: the substitution tracks per-layer coupling at equal parameter count across 1M–108M scales, a frozen $\mathbf{M}$ transfers across datasets with 1.4% validation-loss degradation, and the emergent $\mathbf{M}$ crystallises to a low-rank geometry (effective rank 19–70, coupling-manifold dimensionality 5–8) consistent with what we measure on pretrained transformers.

These are not SOTA claims. Compute-matched ablations, seed-variability error bars, and scale beyond 605M are all future work. What we think is interesting is the shared object itself: low-rank, stable during training, consistent across model families, and directly reusable as a backbone for mixture-of-experts, response-only fine-tuning, and interpretability work that has otherwise had to reason about millions of diffuse per-layer weights. These engineering consequences are where we expect the payoff of the decomposition to lie.

All code, checkpoints, and experiment scripts are publicly available at the repository in the title footnote.

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